捌、附錄

Li

```r
library(showtext)
showtext_auto()  # 啟用 showtext
font_add("Microsoft JhengHei UI", "C:/Windows/Fonts/msjh.ttc")  # 添加你使用的字體
```

載入套件與資料集

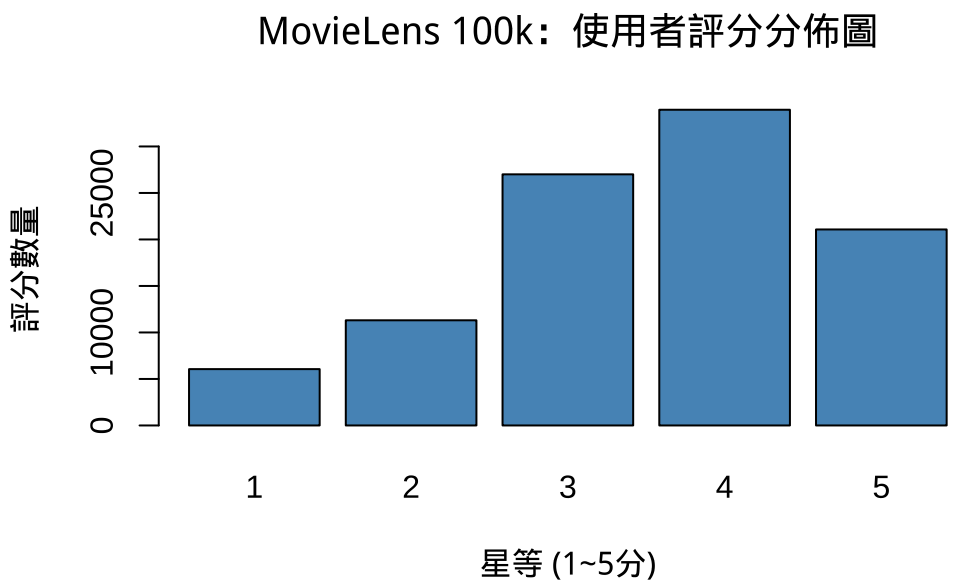
```r
library(ggplot2)
library(tidyr)
library(recommenderlab)
data("MovieLense")
```

壹、探索性資料分析 (EDA)

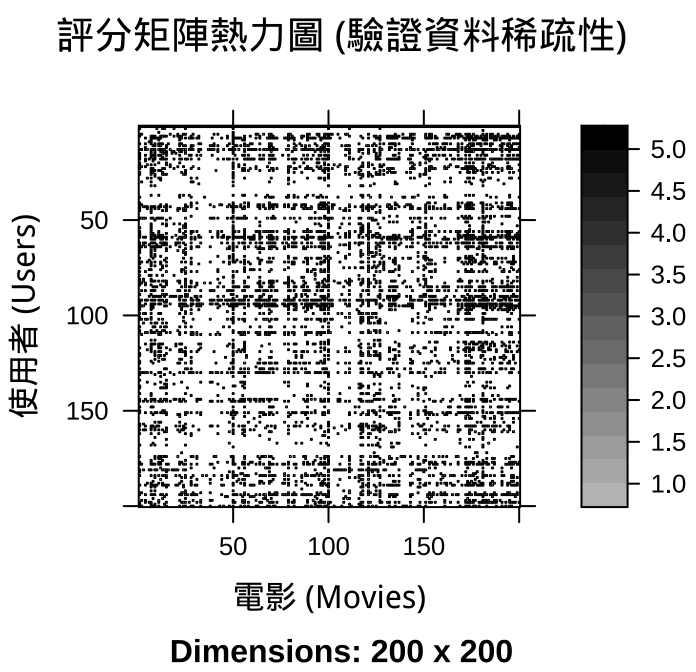
```r
# 1. 檢視資料集維度
print(dim(MovieLense))
```

```
[1]  943 1664
```

```r
# 2. 評分分佈直方圖
ratings_vector <- getRatings(MovieLense)
barplot(table(ratings_vector),
     main = "MovieLens 100k：使用者評分分佈圖",
     xlab = " 星等 (1~5 分)", ylab = " 評分數量",
     col = "steelblue")
```




```
# 3. 評分矩陣熱力圖 (觀察資料稀疏性)
image(MovieLense[1:200,1:200],
      main = " 評分矩陣熱力圖 (驗證資料稀疏性)",
      ylab = " 使用者 (Users)", xlab = " 電影 (Movies)")
```



貳、四種演算法比較

```
# 1. 建立實驗設計 (80% 訓練集，給定 15 筆，>=4 分視為喜歡)
set.seed(2026)
scheme_rating <- evaluationScheme(MovieLense, method="split",
```

```r
                                    train=0.8, given=15, goodRating=4)

# 2. 定義比較模型清單
algorithms <- list(
  "Random"                = list(name="RANDOM", param=NULL),
  "Popular"               = list(name="POPULAR", param=NULL),
  "UBCF"                  = list(name="UBCF",param=list(method="Cosine")),
  "Matrix_Factorization" = list(name="SVDF", param=list(k=10))
)

# 3. 計算預測誤差 (RMSE, MSE, MAE)
results_rating <- evaluate(scheme_rating, algorithms, type="ratings")
```

```
RANDOM run fold/sample [model time/prediction time]
     1  [0.01sec/0.17sec]
POPULAR run fold/sample [model time/prediction time]
     1  [0.3sec/0.12sec]
UBCF run fold/sample [model time/prediction time]
     1  [0.01sec/1.1sec]
SVDF run fold/sample [model time/prediction time]
     1  [149.92sec/19.08sec]
```

```r
# 提取各模型的 RMSE 結果並合併成表格
eval_rmse_results <- do.call(rbind, lapply(results_rating, function(x) avg(x)))
print(eval_rmse_results)
```

```
         RMSE      MSE       MAE
[1,] 1.716589 2.946676 1.4063120
[2,] 1.000568 1.001135 0.7887687
[3,] 1.223650 1.497320 0.9569730
[4,] 1.002266 1.004538 0.7869330
```

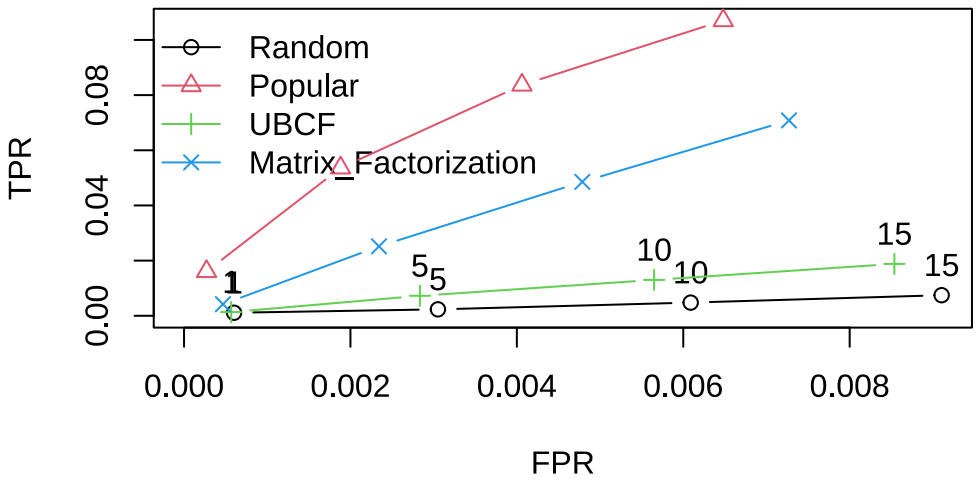
```r
# 4. 計算 Top-N 推薦指標 (Precision, Recall, ROC)
results_topN <- evaluate(scheme_rating, algorithms, n=c(1, 5, 10, 15),
                         type="topNList")
```

```
RANDOM run fold/sample [model time/prediction time]
     1  [0.01sec/0.13sec]
POPULAR run fold/sample [model time/prediction time]
     1  [0.02sec/0.49sec]
UBCF run fold/sample [model time/prediction time]
     1  [0sec/1.89sec]
SVDF run fold/sample [model time/prediction time]
     1  [153.97sec/19.31sec]
```

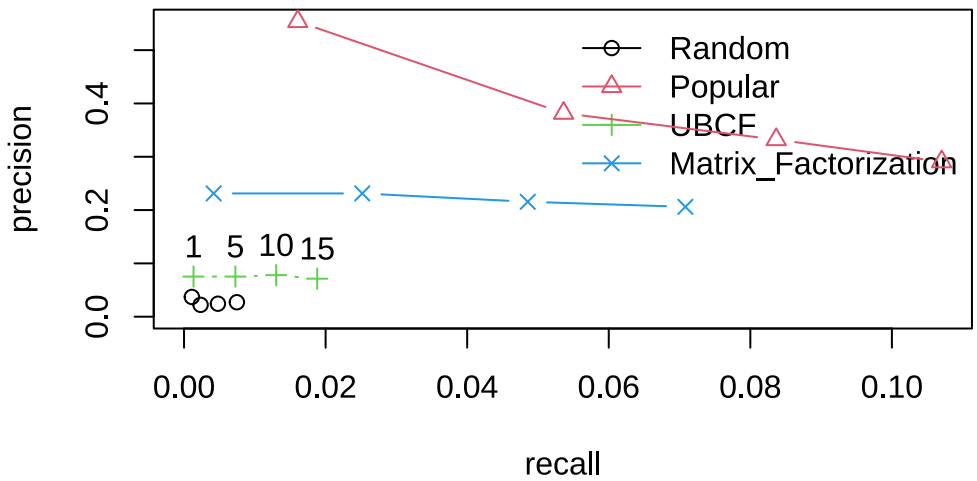
```r
# Top-n 的 Precision 與 Recall 數值 (confusion matrix)
avg(results_topN)
```

```
$Random
             TP        FP       FN       TN        N  precision      recall
[1,] 0.03703704  0.962963 46.86772 1602.254 1650.122 0.03703704 0.001105660
[2,] 0.11111111  4.888889 46.79365 1598.328 1650.122 0.02222222 0.002336261
[3,] 0.24338624  9.756614 46.66138 1593.460 1650.122 0.02433862 0.004784924
[4,] 0.40740741 14.592593 46.49735 1588.624 1650.122 0.02716049 0.007464767
             TPR          FPR  n
[1,] 0.001105660 0.0006010683  1
[2,] 0.002336261 0.0030515720  5
[3,] 0.004784924 0.0060892137 10
[4,] 0.007464767 0.0091054120 15

$Popular
            TP         FP       FN       TN        N precision     recall
[1,] 0.5449735  0.4391534 46.35979 1602.778 1650.122 0.5537634 0.01606764
[2,] 1.8783069  3.0423280 45.02646 1600.175 1650.122 0.3817204 0.05361748
[3,] 3.2698413  6.5714286 43.63492 1596.646 1650.122 0.3322581 0.08365351
[4,] 4.2910053 10.4708995 42.61376 1592.746 1650.122 0.2906810 0.10702648
            TPR          FPR  n
[1,] 0.01606764 0.0002702461  1
[2,] 0.05361748 0.0018818048  5
[3,] 0.08365351 0.0040616772 10
[4,] 0.10702648 0.0064810330 15

$UBCF
             TP         FP       FN       TN        N  precision      recall
[1,] 0.07407407  0.9100529 46.83069 1602.307 1650.122 0.07526882 0.001343220
[2,] 0.37037037  4.5502646 46.53439 1598.667 1650.122 0.07526882 0.007254266
[3,] 0.76719577  9.0740741 46.13757 1594.143 1650.122 0.07795699 0.013012099
[4,] 1.05291005 13.7089947 45.85185 1589.508 1650.122 0.07132616 0.018812805
             TPR          FPR  n
[1,] 0.001343220 0.0005667361  1
[2,] 0.007254266 0.0028347654  5
[3,] 0.013012099 0.0056498802 10
[4,] 0.018812805 0.0085374154 15

$Matrix_Factorization
            TP         FP       FN       TN        N precision     recall
[1,] 0.2275132  0.7566138 46.67725 1602.460 1650.122 0.2311828 0.00418843
[2,] 1.1375661  3.7830688 45.76720 1599.434 1650.122 0.2311828 0.02517352
[3,] 2.1216931  7.7195767 44.78307 1595.497 1650.122 0.2155914 0.04855576
[4,] 3.0423280 11.7195767 43.86243 1591.497 1650.122 0.2060932 0.07082570
            TPR         FPR  n
[1,] 0.00418843 0.000467645  1
[2,] 0.02517352 0.002343396  5
[3,] 0.04855576 0.004786921 10
[4,] 0.07082570 0.007270164 15
```

```r
# 5. 繪製 ROC 與 Precision-Recall 曲線
plot(results_topN, annotate = c(1, 3), legend="topleft", main="ROC 曲線")
```




```
plot(results_topN, "prec/rec", annotate = 3, legend="topright",
     main="Precision-Recall 曲線")
```

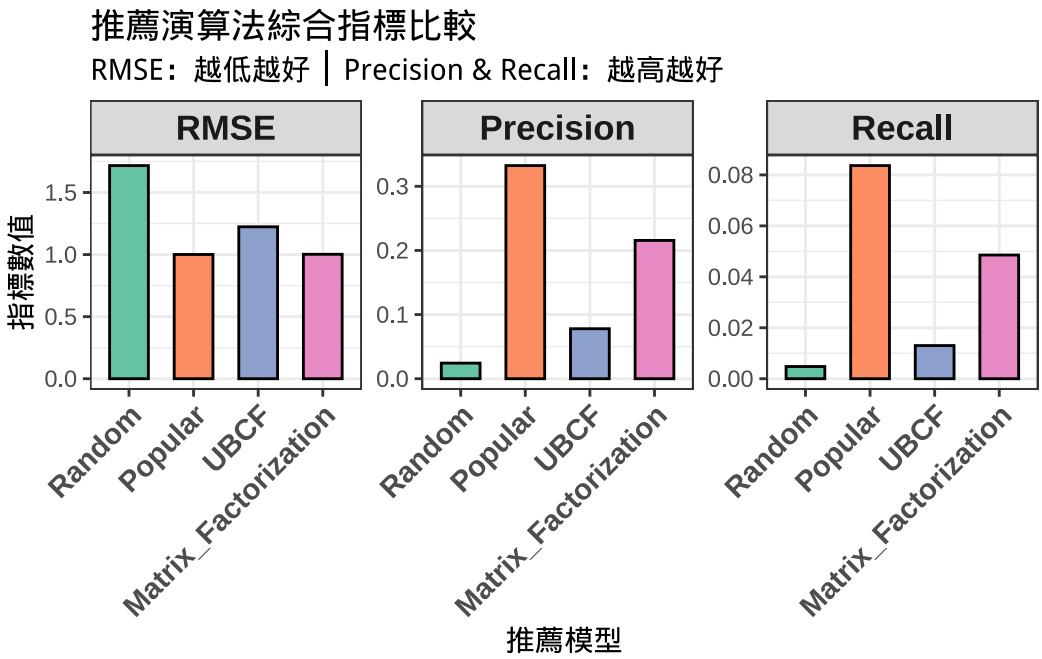


參、綜合比較長條圖 (RMSE, Precision, Recall)

```
# 1. 指定提取 Top-10 的推薦表現
top_n_val <- 3 #(第 3 個是 top-10)

# 2. 整理最終繪圖資料框
final_metrics <- data.frame(
```

```r
  Model = c("Random","Popular", "UBCF", "Matrix_Factorization"),

  RMSE = eval_rmse_results[, "RMSE"],

  Precision = c(
    avg(results_topN)[["Random"]][top_n_val, "precision"],
    avg(results_topN)[["Popular"]][top_n_val, "precision"],
    avg(results_topN)[["UBCF"]][top_n_val, "precision"],
    avg(results_topN)[["Matrix_Factorization"]][top_n_val, "precision"]
  ),

  Recall = c(
    avg(results_topN)[["Random"]][top_n_val, "recall"],
    avg(results_topN)[["Popular"]][top_n_val, "recall"],
    avg(results_topN)[["UBCF"]][top_n_val, "recall"],
    avg(results_topN)[["Matrix_Factorization"]][top_n_val, "recall"]
  )
)
print(final_metrics)
```

```
                  Model     RMSE  Precision      Recall
1                Random 1.716589 0.02433862 0.004784924
2               Popular 1.000568 0.33225806 0.083653514
3                  UBCF 1.223650 0.07795699 0.013012099
4 Matrix_Factorization 1.002266 0.21559140 0.048555757
```

```r
# 3. 資料轉換
metrics_long <- pivot_longer(final_metrics,
                             cols = c(RMSE, Precision, Recall),
                             names_to = "Metric",
                             values_to = "Value")

# 設定 X 軸與分面圖的顯示順序
metrics_long$Model <- factor(metrics_long$Model,
                       levels = c("Random","Popular", "UBCF", "Matrix_Factorization"))
metrics_long$Metric <- factor(metrics_long$Metric,
                              levels = c("RMSE", "Precision", "Recall"))

# 4. 繪製分面長條圖
ggplot(metrics_long, aes(x = Model, y = Value, fill = Model)) +
  geom_bar(stat = "identity", color = "black", width = 0.6) +
  facet_wrap(~ Metric, scales = "free_y") +
  scale_fill_brewer(palette = "Set2") +
  theme_bw() +

  labs(title = " 推薦演算法綜合指標比較",
       subtitle = "RMSE：越低越好   Precision & Recall：越高越好",
       x = " 推薦模型", y = " 指標數值") +
```

```r
  theme(axis.text.x = element_text(angle = 45, hjust = 1, size = 11, face = "bold"),
        strip.text = element_text(size = 13, face = "bold"),
        legend.position = "none")
```